

Experiment Details, Parameters, and Results

1 Experiment Details

This experiment involves cleaning, filtering, and performing sentiment detection on Reddit post data from 9 different subreddits.

- **Input Data Status:** Clean
- **Total Posts Scanned:** 362,063 posts
- **Total Posts Filtered:** 184,045 posts (50.8% of the total)
- **Total Time Elapsed for Filtering:** 22 minutes 46 seconds (average of 151.82 seconds per file)

2 Parameters

2.1 Data Source

Historical Reddit data was fetched from established data collections in computational social science (<https://arctic-shift.photon-reddit.com/download-tool>). Compliance with academic research ethics is ensured, and user privacy is protected.

2.2 Data Source Parameters

The time ranges, post counts, and comment counts configured for each data source are presented in Table 1:

Table 1: Data Source Basic Parameters			
Subreddit	Time Range	Post Count	Comment Count
FC Barca	2021-12-31 to 2026-05-28	72,299	3,550,342
FC Bayern	2019-12-31 to 2026-05-28	17,829	1,220,342
RealMadrid	–	95,641	3,888,310
ManchesterUnited	2021-12-31 to 2026-05-28	73,498	2,199,287
Liverpool FC	2024-12-31 to 2026-05-28	36,601	3,175,480
LaLiga	2020.12.31 to 2026-05-28	9,836	49,308
PremierLeague	2024-12-31 to 2026-05-28	21,142	1,123,524
Bundesliga	2020-12-31 to 2026-05-28	28,060	651,392
Ligue 1 France	2019-12-31 to 2026-5-28	7,157	35,708

2.3 LLM Sentiment Detection Environment

The environment and hardware configurations for the LLM sentiment detection pipeline are specified below:

- **Model Name:** Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4 (14 Billion Params)
- **Framework:** vLLM Boosted
- **GPU:** 1x NVIDIA RTX 4090
- **GPU Memory:** 24GB
- **GPU Memory Utilization:** 92%
- **Max Model Length:** 2048 tokens
- **CUDA Version:** 12.1
- **CPU:** 16x vCPU Intel(R) Xeon(R) Platinum 8358P CPU @ 2.60GHz
- **System RAM:** 90GB
- **Concurrent Concurrency:** 128

3 Results

3.1 Filtering Summary

The statistical results of the filtering process for the 9 post files are shown in Table 2:

Table 2: Data Filtering Statistical Results

Input File	Progress & Speed	Total Posts	Filtered Posts	Ratio	With Player Name	Evaluation Word Only
Barca_cleaned_posts.jsonl	72299 lines [06:14, 193.21 lines/s]	72,299	40,969	56.7%	36,826	4,143
Bundesliga_cleaned_posts.jsonl	28060 lines [01:21, 345.80 lines/s]	28,060	9,105	32.4%	8,061	1,044
LaLiga_cleaned_posts.jsonl	9836 lines [00:28, 351.09 lines/s]	9,836	3,779	38.4%	3,269	510
Ligue1_cleaned_posts.jsonl	7157 lines [00:47, 150.60 lines/s]	7,157	2,311	32.3%	2,149	162
LiverpoolFC_cleaned_posts.jsonl	36601 lines [01:55, 317.40 lines/s]	36,601	19,915	54.4%	17,740	2,175
ManchesterUnited_cleaned_posts.jsonl	73498 lines [03:45, 325.22 lines/s]	73,498	36,503	49.7%	32,469	4,034
PremierLeague_cleaned_posts.jsonl	21142 lines [01:11, 294.06 lines/s]	21,142	9,295	44.0%	7,825	1,470
fcbayern_cleaned_posts.jsonl	17829 lines [01:04, 277.29 lines/s]	17,829	8,196	46.0%	7,068	1,128
realmadrid_cleaned_posts.jsonl	95641 lines [05:57, 267.36 lines/s]	95,641	53,972	56.4%	48,613	5,359

3.2 LLM Sentiment Detection Execution Results

The overall execution results for the LLM sentiment detection pipeline are summarized below:

- **Total Rows Processed:** 184,045 Rows
- **Total Time:** 38,196 Seconds (10.61 Hours)
- **Queries Per Second (QPS):** 4.82